

AI Predictive Issues - Priority Categorization

This document categorizes predictable issues for an AI-powered predictive maintenance platform based on prediction accuracy, business value, and telemetry availability.

■ Super High

Issue	Reason
SSD/HDD Failure Prediction	SMART data enables highly reliable failure prediction.
Battery Lifespan Prediction	Battery degradation follows measurable trends.
Overheating & Thermal Prediction	Continuous temperature monitoring provides early warning.
Fan Failure Prediction	Fan RPM and thermal trends indicate wear.
Performance Degradation	Boot time and resource usage reveal gradual slowdowns.
Remaining Useful Life (RUL)	Combines multiple signals to estimate component lifespan.

■ Very High

Issue	Reason
RAID Failure Prediction	RAID controllers expose detailed health metrics.
Server Health Prediction	Enterprise servers provide rich telemetry.
Application Crash Prediction	Crash logs reveal recurring instability.
Operating System Instability	Event logs often show early warning signs.
Printer Maintenance Prediction	Usage counters accurately forecast maintenance.
Enterprise Fleet Analytics	Fleet-wide data improves AI accuracy.

■ High

Issue	Reason
Network Congestion Prediction	Traffic trends reveal bottlenecks.
Wi-Fi Stability Prediction	Signal quality trends predict issues.
GPU Thermal Prediction	GPU telemetry helps identify overheating.
CPU Thermal Throttling	Load and temperature trends expose throttling.
Disk Performance Bottlenecks	Latency metrics indicate degradation.
Storage Capacity Forecasting	Growth trends enable proactive planning.
Driver Conflict Detection	Driver and event history reveal instability.
Cooling Efficiency Analysis	Thermal data identifies declining cooling.

■ Medium

Issue	Reason
RAM Failure Prediction	Limited telemetry unless ECC memory is present.
PSU Degradation	Consumer power supplies expose limited health data.
Malware Behavior Detection	Behavioral analysis can identify suspicious patterns.
Security Threat Prediction	Requires multiple behavioral indicators.
Network Device Hardware Failure	Depends on available SNMP telemetry.
Thin Client Health Prediction	Limited hardware telemetry.

■ Low

Issue	Reason
Motherboard Failure Prediction	Few reliable warning indicators.
CPU Hardware Failure	Physical CPU failures are rare.
GPU Silicon Failure	Chip failures are difficult to predict.
BIOS Corruption Prediction	Usually caused by sudden events.
Sudden Power Events	External power failures are generally unpredictable.

Recommended Development Roadmap

1. Phase 1 (MVP): SSD/HDD, Battery, Thermal, Fan, Performance, RUL
2. Phase 2: RAID, Servers, Applications, OS, Enterprise Dashboard, Printers
3. Phase 3: Network, GPU, CPU, Drivers, Storage Planning
4. Phase 4: Security, PSU, RAM, Thin Clients
5. Phase 5: Motherboard, CPU Hardware, GPU Chip, BIOS Prediction